

CSC103-Programming Fundamentals

Lecture-5

Instructor

Rizwan Rashid

Lecture Outline

- Variables, Constants
- Standard Output Stream Object (`System.out`)
- Reading Input using the `Scanner` class
- Package, Classes, Methods and `import`

Input

- Storing data in the computer's memory is a two-step process:
 - Instruct the computer to allocate memory
 - Include statements in the program to put the data into to allocated memory

Variables

- A **variable** is a named place in the memory where a programmer can store data (e.g. numbers and letters) and later retrieve the data using the variable “name”
- Data item in a variable is called its **value**
- You can change the contents of a **variable** in a later statement

```
double x;
```

```
int y;
```

```
x = 12.2;
```

```
y = 14;
```

x

12.2

y

14

Variables

- A **variable** is a named place in the memory where a programmer can store data (e.g. numbers and letters) and later retrieve the data using the variable “name”
- Data item in a variable is called its **value**
- You can change the contents of a **variable** in a later statement

```
double x;
```

```
int y;
```

```
x = 12.2;
```

```
y = 14;
```

```
x = 10.2;
```

x

~~12.2~~

10.2

y

14

Variables

Declaration

```
int x;  
double radius;  
char a;
```

Assignment Statement

```
x = 1;  
radius = 1.2;  
a = 'A';
```

Declaration and Assignment

```
int x = 10;  
double radius = 1.2;
```

Identify and fix the errors in the following code

```
1 public class Test {  
2     public static void main(String[] args) {  
3         int i = k + 2;  
4         System.out.println(i);  
5     }  
6 }
```

Named Constants

A named constant is an identifier that represents a permanent value.

- **Syntax**

```
final datatype CONSTANTNAME = value;
```

- **Example**

```
final datatype CONSTANTNAME = VALUE;  
final double PI = 3.14159;  
final int SIZE = 3;
```


Output(System.out)

- Standard output stream object ([System.out](#))

```
System.out.print(expression);  
System.out.println(expression);  
System.out.println();
```

	Statement	Output
1	<code>System.out.println(29 / 4);</code>	7
2	<code>System.out.println("Hello there.");</code>	Hello there.
3	<code>System.out.println(12);</code>	12
4	<code>System.out.println("4 + 7");</code>	4 + 7
5	<code>System.out.println(4 + 7);</code>	11
6	<code>System.out.println('A');</code>	A
7	<code>System.out.println("4 + 7 = " + (4 + 7));</code>	4 + 7 = 11
8	<code>System.out.println(2 + 3 * 5);</code>	17
9	<code>System.out.println("Hello \nthere.");</code>	Hello there.

Reading Input from the Console

Reading input from the console enables the program to accept input from the user.

1. Create a Scanner object

`System.in` - standard input stream object

```
Scanner input = new Scanner(System.in);
```

2. Use the method **nextDouble()** to obtain to a double value. For example,

```
System.out.println("Enter the radius of Circle");  
double radius = input.nextDouble();
```

Reading Numbers from the Keyboard

```
Scanner input = new Scanner(System.in) ;  
int value = input.nextInt() ;
```

Method	Description
<code>nextByte()</code>	reads an integer of the <code>byte</code> type.
<code>nextShort()</code>	reads an integer of the <code>short</code> type.
<code>nextInt()</code>	reads an integer of the <code>int</code> type.
<code>nextLong()</code>	reads an integer of the <code>long</code> type.
<code>nextFloat()</code>	reads a number of the <code>float</code> type.
<code>nextDouble()</code>	reads a number of the <code>double</code> type.

Reading Numbers from the Keyboard

```
Scanner input = new Scanner(System.in);  
System.out.print("Enter a byte value: ");  
byte byteValue = input.nextByte();
```

```
System.out.print("Enter a short value: ");  
short shortValue = input.nextShort();
```

```
System.out.print("Enter an int value: ");  
int intValue = input.nextInt();
```

```
System.out.print("Enter a long value: ");  
long longValue = input.nextLong();
```

```
System.out.print("Enter a float value: ");  
float floatValue = input.nextFloat();
```

Reading a character

```
Scanner input = new Scanner(System.in);
```

`input.next()` : get the next input token as string

`input.nextLine()` : get the next input as a string until the end of line

```
char ch = input.next().charAt(0);
```

Package, Classes, Methods and `import`

- A package is a collection of related classes
 - `java.lang` package contain many classes e.g. `Math`, `System`
 - `java.util` package contain `Scanner` class

Specific Import:
`import java.util.Scanner;`

Wildcard Import:
`import java.util.*;`

- `class` is used to create Java programs
 - Use to group set of related operations(methods)

```
import java.util.Scanner; // Scanner is in the java.util package
```

```
public class ComputeArea {  
    public static void main(String[] args){
```

```
        // Create a Scanner object  
        Scanner input = new Scanner(System.in);
```

```
        // Prompt the user to enter a radius  
        System.out.println("Enter the radius of Circle");  
        double radius = input.nextDouble();
```

```
        // Compute area  
        double area = 3.14159 * radius * radius;
```

```
        // Display results  
        System.out.print("The area of Circle is: "+area);
```

```
    }
```

```
}
```